BOILER & STEAM PIPING

Efficient Steam Generation, Safe Distribution, Proper Utilization

1. BOILER – BASICS

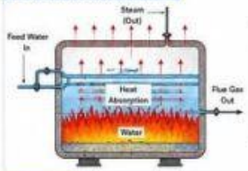
A boiler is a closed vessel in which water is heated and converted into steam at a required pressure.

Main Parts

1. Shell / Drum
2. Furnace
3. Water Tubes / Fire Tubes
4. Burner
5. Superheater
6. Safety Valve
7. Steam Stop Valve
8. Water Level Gauge
9. Feed Check Valve
10. Blowdown Valve



2. STEAM GENERATION PROCESS



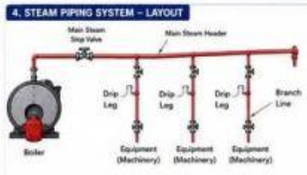
Heat from the furnace transfers to water through the walls. Water boils and changes into steam.

Steam collects at the top and is sent to steam mains.

3. STEAM PROPERTIES (SATURATED STEAM)

Pressure (kg/cm²(g))	Saturation Temperature (°C)	Specific Enthalpy (kJ/kg)	Specific Volume (m³/kg)
2	120.2	2706	0.8853
4	143.6	2728	0.4624
7	165.0	2748	0.2720
10	179.9	2776	0.1944
14	195.0	2798	0.1450

4. STEAM PIPING SYSTEM – LAYOUT



Typical Piping Arrangement

1. Steam rises up from boiler outlet.
2. Main steam stop valve.
3. Main steam header sloped in direction of flow.
4. Drip legs at each take-off point.
5. Branch lines to equipment.
6. Drain / steam traps to return condensate.

5. STEAM PIPING – GENERAL GUIDELINES

- Use proper size pipe to avoid pressure drop.
- Steam mains should be sloped in the direction of steam flow (1 : 200 to 1 : 100).
- Provide drip legs at all low points and branch connections.
- Insulate all steam pipes to prevent heat loss and condensation.
- Use expansion loops or joints for long runs.
- Install steam traps at the lowest point of equipment.

6. MAIN STEAM PIPE SIZING (RECOMMENDED)

Recommended Steam Velocities

Pressure (kg/cm²(g))	Recommended Velocity (m/s)
Up to 4	20 – 25
4 to 10	25 – 30
10 to 17	30 – 35
Above 17	30 – 40

Pipe Size Calculation Formula

$$d = \sqrt{\frac{4Q}{\pi \times v \times 3600}}$$

Where,

d = Internal diameter of pipe (m)

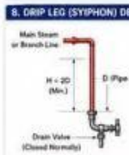
Q = Steam flow (kg/h)

v = Velocity of steam (m/s)

7. PIPE MATERIAL

- Carbon Steel (Seamless) – Most common for steam piping.
- Stainless Steel – For corrosive or high temperature service.
- Copper / Copper Alloy – For low pressure steam.

8. DRIP LEG (SYPHON) DETAILS



Purpose

To collect and remove moisture and prevent it from entering the equipment.

Design

- Height (H) = 2 x Pipe Size (Min.)
- Provide drain valve at bottom.
- Insulate the drip leg.

9. STEAM TRAP



Function

Automatically removes condensate, air and non-condensable gases without allowing steam to escape.

Common Types

- Float & Thermostatic Trap
- Thermodynamic Trap
- Inverted Bucket Trap
- Bimetallic Trap

10. PIPE SUPPORT & HANGERS

- Provide proper supports to carry the weight of pipe, insulation and steam.
- Allow for thermal expansion.
- Use rigid hangers near valves and equipment.
- Use spring hangers or guides for long horizontal runs.



11. INSULATION THICKNESS (RECOMMENDED)

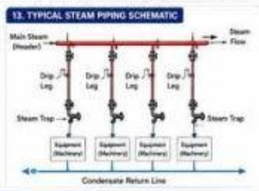
Steam Pipe Size (NB)	Insulation Thickness (mm) Up to 7 kg/cm²(g)	Above 7 kg/cm²(g)
15 – 25	25	38
32 – 50	38	50
65 – 100	50	65
125 – 150	65	75
200 & Above	75	100

Use good quality insulation (Glass Wool / Rock Wool) with weather protection.

12. STEAM PIPING DO's & DON'Ts

DO's	DON'Ts
Slope the mains in direction of flow.	Do not allow water hammer.
Install drip legs at all low points.	Do not reduce pipe size suddenly.
Insulate all steam pipes.	Do not share steam trap discharge line with others.
Use proper size pipe and fittings.	Do not put strain on connected equipment.
Test the system for leaks.	Do not block or cap drip legs.
Maintain steam traps regularly.	

13. TYPICAL STEAM PIPING SCHEMATIC



14. SAFETY PRECAUTIONS

- Ensure safety valve is in good condition.
- Do not exceed design pressure.
- Check water level regularly.
- Blowdown boiler as per schedule.
- Wear PPE and follow safe work practices.

WARNING

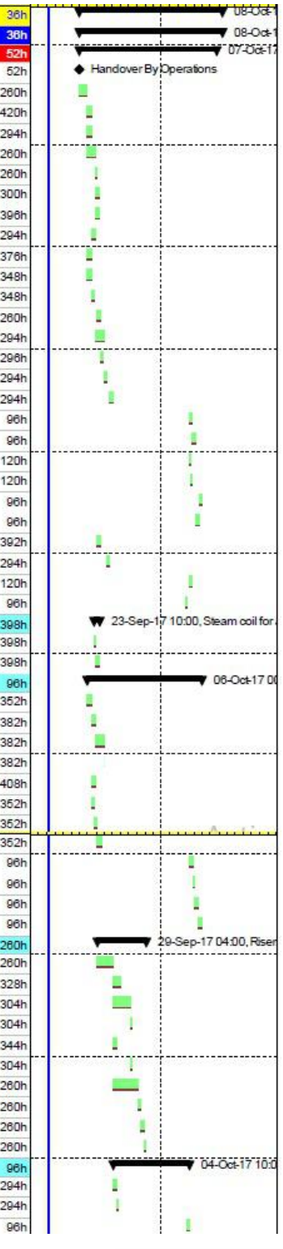
Steam is extremely hot and under pressure. Improper handling can cause severe injury or death.

15. BENEFITS OF PROPER STEAM PIPING





STEAM GENERATOR		420h		21-Sep-17 00:00	08-Oct-17 12:00			36h
SRU Train A & B		420h		21-Sep-17 00:00	08-Oct-17 12:00			36h
Waste Heat Boiler SRU Train A		404h		21-Sep-17 00:00	07-Oct-17 20:00			52h
129E101	Handover By Operations	0h	0%	21-Sep-17 00:00				52h
STG2000	Spading	24h	0%	21-Sep-17 00:00	22-Sep-17 00:00			260h
STG2010	Connect Steam, Air Hoses for depressuri	12h	0%	22-Sep-17 00:00	22-Sep-17 12:00			420h
STG2020	Erection of external scaffolding	12h	0%	22-Sep-17 00:00	22-Sep-17 12:00			294h
STG2030	Open all manways	24h	0%	22-Sep-17 00:00	23-Sep-17 00:00			260h
STG2040	Obtain Entry permit	4h	0%	23-Sep-17 00:00	23-Sep-17 04:00			260h
STG2050	External cleaning of tubes	12h	0%	23-Sep-17 00:00	23-Sep-17 12:00			300h
STG2060	Remove ceramic ferrules	12h	0%	23-Sep-17 00:00	23-Sep-17 12:00			398h
STG2070	Removal of external insulation	12h	0%	22-Sep-17 12:00	23-Sep-17 00:00			294h
STG2080	Open all hand hole blind flanges of shell	12h	0%	22-Sep-17 00:00	22-Sep-17 12:00			376h
STG2090	Drop all 1 stisolation valves & offer hydrote	12h	0%	22-Sep-17 00:00	22-Sep-17 12:00			348h
STG2100	Deplug all the nozzles and offer for inspec	8h	0%	22-Sep-17 12:00	22-Sep-17 20:00			348h
STG2110	Internal Cleaning of tubes	12h	0%	23-Sep-17 04:00	23-Sep-17 16:00			260h
STG2120	External Surfaoe cleaning of Boiler syster	24h	0%	23-Sep-17 00:00	24-Sep-17 00:00			294h
STG2130	UT guaging	6h	0%	23-Sep-17 16:00	23-Sep-17 22:00			298h
STG2140	Dry inspection by BPCL	6h	0%	24-Sep-17 00:00	24-Sep-17 06:00			294h
STG2150	Dry Inspection by GBI	8h	0%	24-Sep-17 18:00	25-Sep-17 02:00			294h
STG2190	Removal of internal scaffolding	8h	0%	04-Oct-17 10:00	04-Oct-17 18:00			96h
STG2200	Box up Manways	12h	0%	04-Oct-17 18:00	05-Oct-17 06:00			96h
STG2210	Application of external paint	4h	0%	04-Oct-17 10:00	04-Oct-17 14:00			120h
STG2220	Application of insulation pocket	4h	0%	04-Oct-17 14:00	04-Oct-17 18:00			120h
STG2230	Removal of external scaffolding	6h	0%	05-Oct-17 18:00	06-Oct-17 00:00			96h
STG2240	Despading	12h	0%	05-Oct-17 06:00	05-Oct-17 18:00			96h
STG2250	Erection of Internal Scaffolding	12h	0%	23-Sep-17 04:00	23-Sep-17 16:00			392h
STG2260	Insert ceramic ferrules	12h	0%	24-Sep-17 06:00	24-Sep-17 18:00			294h
STG2270	Box up all hand holes	8h	0%	04-Oct-17 10:00	04-Oct-17 18:00			120h
STG2280	Hydrotest preparation for GBI	4h	0%	04-Oct-17 00:00	04-Oct-17 04:00			96h
Steam coil for all Sulphur Condensors		16h		22-Sep-17 18:00	23-Sep-17 10:00			398h
A106	Clean the steam coils and offer for inspec	8h	0%	22-Sep-17 18:00	23-Sep-17 02:00			398h
A106	Offer Hydrotest of steam coils	8h	0%	23-Sep-17 02:00	23-Sep-17 10:00			398h
Steam Drums of 129-E-101		336h		22-Sep-17 00:00	06-Oct-17 00:00			96h
STG2	Open the drum man ways	12h	0%	22-Sep-17 00:00	22-Sep-17 12:00			352h
STG2	Remove all the drum internals	12h	0%	22-Sep-17 12:00	23-Sep-17 00:00			382h
STG2	Clean drum internally and internals	24h	0%	23-Sep-17 00:00	24-Sep-17 00:00			382h
STG2	Offer the drum and drum internals for BP(	2h	0%	24-Sep-17 00:00	24-Sep-17 02:00			382h
STG2	Drop all the first isolation valves and offer l	12h	0%	22-Sep-17 12:00	23-Sep-17 00:00			408h
STG2	Remove all the mountings of steam drum	6h	0%	22-Sep-17 12:00	22-Sep-17 18:00			352h
STG2	Open insulation pockets at all nozzles,she	12h	0%	22-Sep-17 18:00	23-Sep-17 06:00			352h
STG2	Offer the boiler system to GBI for dry inspe	12h	0%	23-Sep-17 06:00	23-Sep-17 18:00			352h
STG2	Surface preparation and paint the drum ir	12h	0%	04-Oct-17 10:00	04-Oct-17 22:00			96h
STG2	Fix back all the internals	2h	0%	04-Oct-17 22:00	05-Oct-17 00:00			96h
STG2	Box up drum manways	12h	0%	05-Oct-17 00:00	05-Oct-17 12:00			96h
STG2	Install all the mountings	12h	0%	05-Oct-17 12:00	06-Oct-17 00:00			96h
Risers/Downcomers/Blowdown Lines for 129-E-10		144h		23-Sep-17 04:00	29-Sep-17 04:00			260h
STG6	Provide External Scaffolding	48h	0%	23-Sep-17 04:00	25-Sep-17 04:00			260h
STG6	Open insulation pockets at all elbows	24h	0%	25-Sep-17 04:00	26-Sep-17 04:00			328h
STG6	Open insulation pockets in straight sectior	48h	0%	25-Sep-17 04:00	27-Sep-17 04:00			304h
STG6	inspection of elbows/downcomers/risers	4h	0%	27-Sep-17 04:00	27-Sep-17 08:00			304h
STG6	Carry out scrape cleaning & painting	8h	0%	25-Sep-17 04:00	25-Sep-17 12:00			344h
STG6	carry out insulation	4h	0%	27-Sep-17 04:00	27-Sep-17 08:00			304h
STG6	De lag the blow down line & offer for inspe	72h	0%	25-Sep-17 04:00	28-Sep-17 04:00			260h
STG6	Hydrotest of blow down line & offer for insp	8h	0%	28-Sep-17 04:00	28-Sep-17 12:00			260h
STG6	Surface preparation and Painting of the li	8h	0%	28-Sep-17 12:00	28-Sep-17 20:00			260h
STG6	Re lag the blow down lines	8h	0%	28-Sep-17 20:00	29-Sep-17 04:00			260h
Hydrotesting of Boilers E-101		224h		25-Sep-17 02:00	04-Oct-17 10:00			96h
STG2	Preparation of hydrotest	12h	0%	25-Sep-17 02:00	25-Sep-17 14:00			294h
STG2	Hydrotest witness by BPCL	4h	0%	25-Sep-17 14:00	25-Sep-17 18:00			294h
STG2	Hydrotest witness by GBI	6h	0%	04-Oct-17 04:00	04-Oct-17 10:00			96h



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